

Functions

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    salute = "Hello"  
    return salute + " " + name
```

```
message = greeting("Mogens")  
print(message)
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“Temporary mini-world”

➔

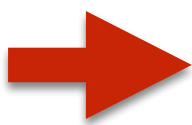
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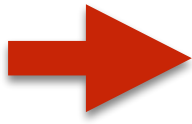


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```
print(message)
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Miniopgave

Skriv en funktion der returnerer kvadratet af et tal:

```
square(4)
```

(skal returnere 16)



Miniopgave

Skriv en funktion der returnerer summen af kvadrater
(sum of squares)

```
sum_of_squares(4, -2, 3)
```

(skal returnere 29)

```
def square(v):  
    return v**2
```

```
def sum_of_squares(a, b, c):  
    a_sq = square(a)  
    b_sq = square(b)  
    c_sq = square(c)  
    return a_sq + b_sq + c_sq
```

```
ssq = sum_of_squares(4, -2, 3)  
print(ssq)
```

```
def square(v):  
    return v**2
```

```
def sum_of_squares(a, b, c):  
    return square(a) + square(b) + square(c)
```

```
ssq = sum_of_squares(4, -2, 3)  
print(ssq)
```

Funktionens kaldets midlertidige og private verden

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def greeting(name):  
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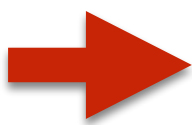
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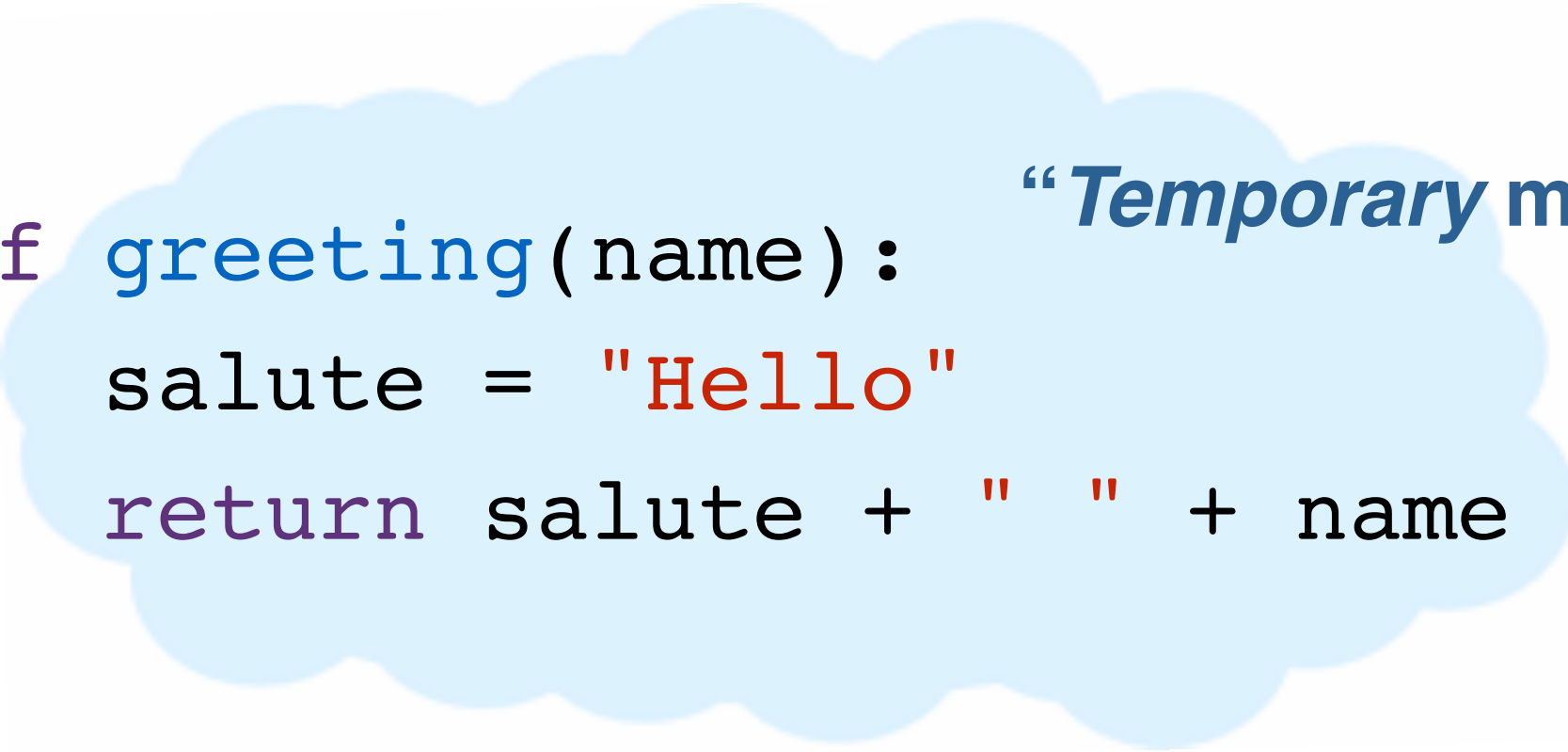


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“Temporary mini-world”

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Funktionens kaldets midlertidige og private verden



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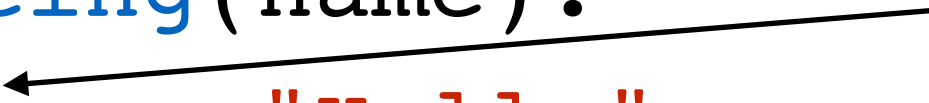


```
print(message)
```

Funktionens private midlertidige verden

```
def greeting(name):  
    salute = "Hello"  
    return salute + " " + name
```

salute is private
to the function



```
message = greeting("Mogens")  
print(message)
```


Funktionens private midlertidige verden

```
def greeting(name):  
    salute = "Hello"  
    return salute + " " + name
```

salute is *still*
private to the
function

```
salute = "Bonjour"  
message = greeting("Mogens")  
print(message)  
print(salute)
```


Funktionens private midlertidige verden

```
def greeting(name):  
    # salute = "Hello"  
    return salute + " " + name
```

Python has to look for
salute outside the
function. Does it find it?

```
salute = "Bonjour"  
message = greeting("Mogens")  
  
print(message)
```

Funktionens private midlertidige verden

```
def greeting(name):  
    # salute = "Hello"  
    return salute + " " + name
```

Python has to look for
salute outside the
function. Does it find it?

```
message = greeting("Mogens")  
salute = "Bonjour"  
print(message)
```

Variable og funktioner

- Et funktionskald skaber en midlertidig verden som forsvinder igen når funktionen returnerer en værdi.
- Alle variable man giver værdier i en funktion er private for funktionen.
- Hvis en variabel ikke er defineret i en funktion vil Python prøve at finde en definition udenfor funktionen.

Hvad er galt?

```
def greeting(name):  
    name = ""  
    salute = "Hello"  
    return salute + " " + name  
  
message = greeting("Mogens")  
print(message)
```

Hvad er galt?

```
def greeting(nama):  
    salute = "Hello"  
    return salute + " " + name
```

```
name = "Henning"  
message = greeting("Mogens")  
print(message)
```



Builtin functions

```
abs_val = abs(-42)  
print(abs_val)
```

```
fruit = "Banana"  
length = len(fruit)  
print(length)
```

```
print(type(length))  
print(type(fruit))
```

<https://docs.python.org/3.8/library/functions.html>



Python 3.9 dokumentation

<https://docs.python.org/3.9/>

Look for "Library Reference"



Miniopgave

Skriv en funktion der tester om et tal er lige eller ulige

```
is_even(42)
```

skal returnere **True**

```
is_even(17)
```

skal returnere **False**